Description

Background

Both in China and worldwide, the incidence of cancer is on the rise, posing a severe threat to human health and claiming lives at an alarming rate. Epidemiological data show that China's lifetime cancer incidence risk is about 27.6%, slightly higher than the global average of 25.1%—meaning roughly 1 in 4 people around the world face the threat of cancer in their lifetime. In terms of point prevalence rate, China reaches about 0.34% while the global rate is 0.67%, which fully highlights the urgent demand for effective cancer prevention, diagnosis and treatment strategies globally.

Among all malignant tumors, lung cancer remains the leading cause of cancer-related deaths, bringing heavy physical, psychological and economic burdens to patients, families and society. As a highly aggressive subtype of lung cancer, Small Cell Lung Cancer (SCLC) is a high-grade neuroendocrine carcinoma that has become one of the most intractable types of cancer to treat, mainly due to its high malignancy, rapid progression and tendency of early metastasis. Cigarette smoking is the primary risk factor for SCLC, with 95% of diagnosed patients having a history of tobacco use.

SCLC has distinct epidemiological and clinical characteristics that distinguish it from other lung cancer subtypes. In 2021, the incidence of SCLC in the United States was 4.7 cases per 100,000 individuals, and its 5-year overall survival rate is only 12% to 30%, far lower than that of non-small cell lung cancer. Clinically, SCLC is classified into limited stage (LS-SCLC, accounting for 30%) and extensive stage (ES-SCLC, accounting for 70%), depending on whether the disease can be covered by a radiation field usually confined to one hemithorax (which may include contralateral mediastinal and supraclavicular nodes). Notably, approximately 15% of patients have brain metastases at the time of diagnosis, further worsening the prognosis.

Despite decades of research, current treatments for SCLC still have obvious limitations and cannot meet clinical needs. The mainstream first-line treatment is a combination of platinum-etoposide chemotherapy and immunotherapy, which achieves an initial tumor shrinkage rate of 60% to 70%, but the therapeutic effect is unsustainable. Patients with ES-SCLC treated with this regimen have a median overall survival of only about 12 to 13 months, and 60% of them experience recurrence within 3 months, accompanied by prominent drug resistance problems.

Second-line therapies for ES-SCLC also have limited efficacy. Commonly used options include the DNA-alkylating agent lurbinectedin, which has an overall response rate of 35% and a median progression-free survival of 3.7 months, and tarlatamab, a bispecific T-cell engager targeting delta-like ligand 3, with an overall response rate of 40% and a median progression-free survival of 4.9 months—neither of which can significantly improve the long-term survival of patients. In addition, SCLC is difficult to screen early due to its aggressive clinical course and early metastasis; the limited use of clinical biomarkers, which is affected by genetic and epigenetic changes during treatment and the difficulty in obtaining high-quality tissue samples, further traps its treatment in a dilemma of "difficult diagnosis, difficult cure, high recurrence rate and high cost".

In summary, the increasing global cancer burden, the high mortality of lung cancer, and the unique aggressiveness and poor prognosis of SCLC, as well as the obvious shortcomings of current treatment methods in terms of efficacy and clinical application, indicate an urgent need for in-depth research on SCLC to develop more effective, safe and economical diagnosis and treatment strategies.

Project Overview

To address these unmet needs, our project devevelops an innovative way of killing cancer cells by using specially designed peptides to trigger protein aggregation in the cells. We takes *Escherichia coli Nissle 1917* (EcN) as the core chassis, leveraging its tumor-tropic and pulmonary compatibility advantages, and designs modular circuits including targeted delivery, pulmonary retention, precision targeting and stimuli-responsive therapy to achieve safe, specific and effective SCLC treatment while maintaining pulmonary homeostasis.

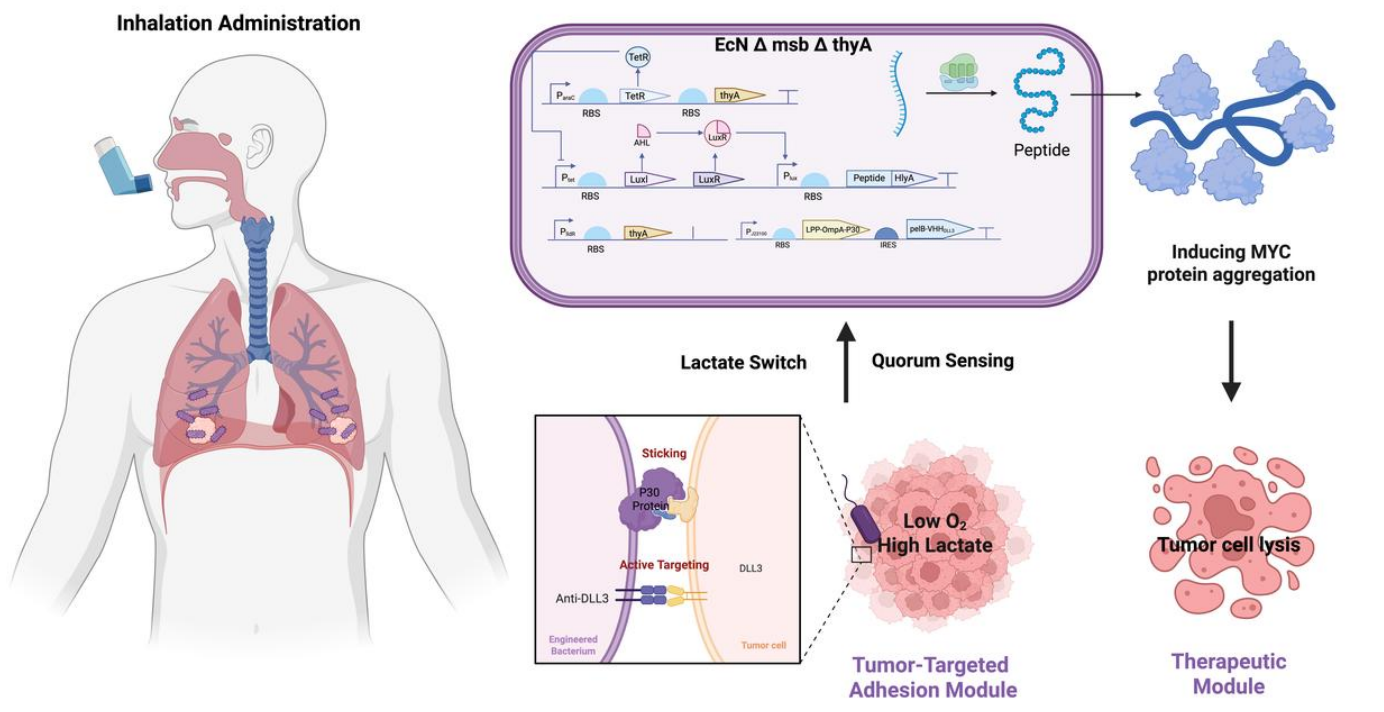

Fig. 1 Schematic illustration of Engineered Bacteria Therapy

Chassis Selection: *Escherichia coli* Nissle 1917 (EcN)

Our project utilizes EcN as the core chassis. As a clinically validated probiotic, EcN offers an exemplary safety profile and high human compatibility. Crucially, EcN exhibits **innate tumor-tropic motility** , autonomously migrating toward the hypoxic and high-lactate TME characteristic of SCLC. Since *E. coli* is a constituent of the native pulmonary microbiota, our engineered EcN is optimized to maintain **ecological homeostasis** while colonizing the respiratory tract.

Engineering Strategy and Modular Circuits

1. Targeted Pulmonary Delivery via Aerosolization

Nebulization delivers engineered bacteria directly to the lung, with **aerosol particle size** optimized for deposition in SCLC-affected bronchial and distal airways. The bacteria exploit the SCLC microenvironment’ s hypoxia and high lactate as metabolic cues for preferential colonization, and surface-expressed **DLL3 nanobodies** enable active recognition of SCLC cells. This multi-layered strategy maximizes intrapulmonary bacterial concentration and minimizes systemic exposure.

2. Pulmonary Colonization Module: Bronchial Adhesion via P30

Engineered bacteria heterologously express **Mycoplasma pneumoniae P30 adhesin** on their surface, anchoring to the bronchial epithelium. This resists mucociliary clearance and establishes stable colonization at tumor-proximal sites for prolonged pulmonary retention.

3. Density-Dependent Activation & Homeostatic Control (Quorum Sensing Circuit)

A **quorum sensing (QS) module** regulates bacterial population and therapy: QS activation at threshold density triggers therapeutic short peptide secretion. This maintains **population homeostasis** and avoids excessive immune stimulation or overgrowth.

4.Metabolic Microenvironment Sensing Module (Lactate-Responsive Circuit)

A lactate-sensing switch controls engineered bacteria survival: leveraging SCLC’ s Warburg effect (high lactate), the module permits bacterial survival and colonization only in **lactate-rich tumor microenvironments** (TME), while restricting viability in low-lactate normal tissues—laying the foundation for QS-mediated peptide secretion.

5. Therapeutic Effector Module: Hemolysin-Mediated Secretion & Amyloid-Induced Cytotoxicity

We design FuncPept by identifying N-MYC **aggregation-prone regions (APRs)** with TANGO and validating via WALZE, supported by integrated software. Legumain-cleavable shielding sequences activate FuncPept only in acidic SCLC lysosomes, avoiding off-target effects. The **HlyA Type I secretion system** directly exports FuncPept into the TME. Tuftsin is conjugated to FuncPept to enhance macrophage-mediated amyloid clearance, while inhalable, nanocarrier-encapsulated **IFN-γ** reprograms TAMs to anti-tumor M1 phenotypes, boosting immunity and reshaping the TME for enhanced efficacy

6. Biosafety Module: Programmed Suicide Switch

ΔmsbB-engineered bacteria lack the endogenous thyA gene (thymidylate synthase). **In vitro**, arabinose induces thyA expression via the ParaC promoter, supporting bacterial expansion. **In vivo**, thyA expression is controlled by the lactate-responsive PlldR promoter, restricting bacterial survival to the high-lactate SCLC TME. This strict auxotrophy ensures controlled bacterial clearance in vivo and ex vivo, meeting stringent biosafety standards.

References

- Cao, W. et al. Cancer statistics in China: Epidemiology, risk factors, and prevention. *Chin. J. Cancer Res.* **37**, 912–928 (2025).
- Bray, F. et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA. Cancer J. Clin.* **74**, 229–263 (2024).
- Kim, S. Y., Park, H. S. & Chiang, A. C. Small Cell Lung Cancer: A Review. *JAMA* **333**, 1906–1917 (2025).
- Thomas, A., Mohindroo, C. & Giaccone, G. Advancing therapeutics in small-cell lung cancer. *Nat. Cancer* **6**, 938–953 (2025).
- Kanak, B. et al. Engineering *Escherichia coli* Nissle 1917 for the delivery of therapeutic proteins. *Trends Pharmacol. Sci.***44**, 761–773 (2023).
- Steidler, L. et al. Biological containment of genetically modified *Lactococcus lactis* for intestinal delivery of human interleukin 10. *Nat. Biotechnol.* **21**, 785–789 (2003).
- Toso, J. F. et al. Phase I study of the intravenous administration of attenuated *Salmonella typhimurium* to patients with metastatic melanoma. *J. Clin. Oncol.* **20**, 142–152 (2002).